

COURSE NAME & CODE: ECA 0303 – Signals and systems

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Experiment 6.1

Convolution of Continuous Time Signals

Aim:

To perform the convolution of two continuous-time signals using Scilab. The Aim is to understand the concept of convolution as a mathematical operation that combines two signals to produce a third signal, representing the amount of overlap between the two as one signal is shifted over the other.

Program:

% Test Case: Convolution of Continuous-Time Signals

% Define time range and sampling interval

t_min = -5; % Start time

t_max = 5; % End time

dt = 0.01; % Time step

t = t_min:dt:t_max; % Time vector

% Define Signal 1: Rectangular pulse of width 2 centered at t = 0

x_t = (t >= -1) .* (t <= 1);

% Define Signal 2: Exponential decay function

h_t = exp(-t) .* (t >= 0); % Exponential decay, causal

% Perform the convolution

```
y_t = conv(x_t, h_t) * dt; % Multiply by dt to scale correctly
```

```
% Adjust the time vector for the convolution result
```

```
t_conv = (2 * t_min):dt:(2 * t_max);
```

```
% Plot Signal 1
```

```
clf; % Clear the current figure
```

```
subplot(3,1,1);
```

```
plot(t, x_t, 'b');
```

```
xlabel('t');
```

```
ylabel('x(t)');
```

```
title('Signal 1: Rectangular Pulse');
```

```
% Plot Signal 2
```

```
subplot(3,1,2);
```

```
plot(t, h_t, 'g');
```

```
xlabel('t');
```

```
ylabel('h(t)');
```

```
title('Signal 2: Exponential Decay');
```

```
% Plot the Convolution Result
```

```
subplot(3,1,3);
```

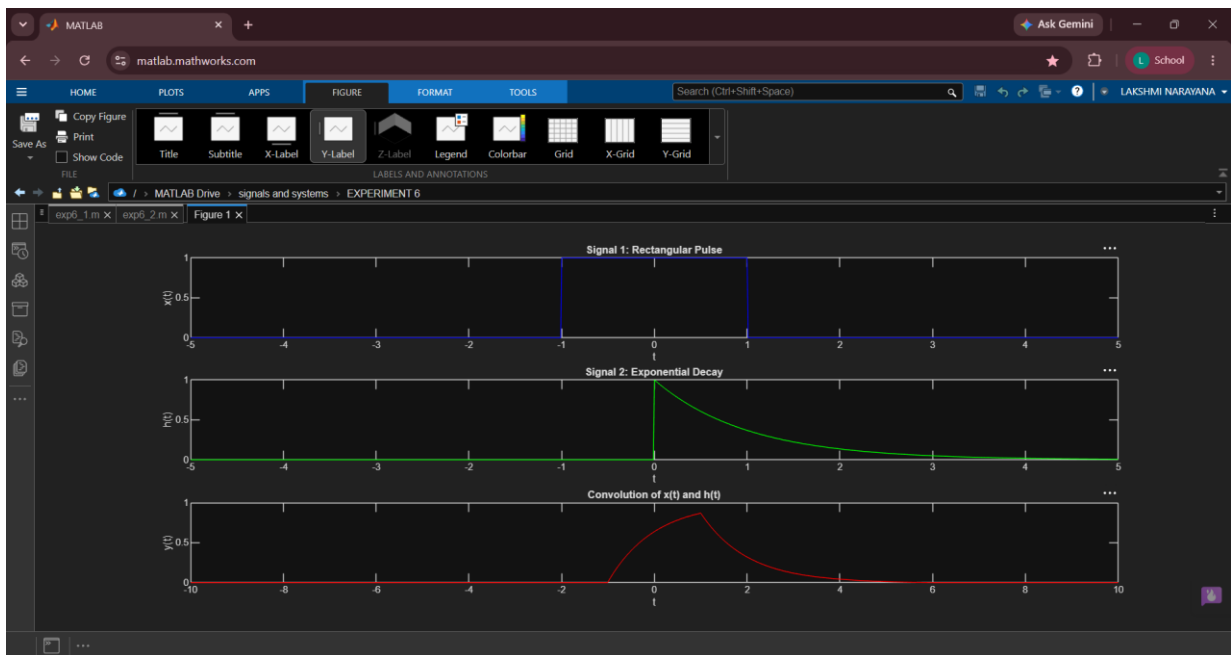
```
plot(t_conv, y_t, 'r');
```

```
xlabel('t');
```

```
ylabel('y(t)');
```

```
title('Convolution of x(t) and h(t)');
```

Output:



Result:

- **Convolution Operation:** Convolution combines the two signals by integrating the product of one signal with a time-shifted version of the other. This

operation is used to analyze the output of a linear time-invariant (LTI) system when given an input signal.

- **Result Interpretation:** The convolution of the rectangular pulse with the exponential decay function results in a signal that resembles the exponential decay but modified by the duration of the pulse. This result illustrates how the convolution operation can be used to model the response of a system (exponential decay) to an input (rectangular pulse).

This test case demonstrates the use of convolution in analyzing continuous-time

signals, showing how one signal is affected by another when passed through a linear time-invariant system.